

Identification of select glucocorticoids as Smoothened agonists: Potential utility for regenerative medicine

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Regenerative medicine holds the promise of replacing damaged tissues largely by stem cell activation. Hedgehog signaling through the plasma membrane receptor Smoothened (Smo) is an important process for regulating stem cell proliferation. The development of Hedgehog-related therapies has been impeded by a lack of US Food and Drug Administration (FDA)-approved Smo agonists. Using a high-content screen with cells expressing Smo receptors and a β -arrestin2-GFP reporter, we identified four FDA-approved drugs, halcinonide, fluticasone, clobetasol, and fluocinonide, as Smo agonists that activate Hedgehog signaling. These drugs demonstrated an ability to bind Smo, promote Smo internalization, activate Gli, and stimulate the proliferation of primary neuronal precursor cells alone and synergistically in the presence of Sonic Hedgehog protein. Halcinonide, fluticasone, clobetasol, and fluocinonide provide an unprecedented opportunity to develop unique clinical strategies to treat Hedgehog-dependent illnesses.

steroids | Hedgehog signaling | Gli | stem cell proliferation | arrestin

The Hedgehog signaling pathway, mediated by the Smoothened (Smo) receptor, has been shown to regulate stem cells and is a fundamental regulator of organogenesis in developing embryos and tissue integrity in mature organisms (1–9). Smo agonists have been proposed as desired therapeutics for restoring tissue function in diseases associated with heart failure, neuronal injury/degeneration, wound repair, and retinal damage, where they could reactivate or stimulate repair mechanisms in situations in which normal regenerative capacity is compromised (10–14). However, for therapeutic modalities to be acceptable for a previously undescribed use in humans, clinical safety and efficacy must be demonstrated to gain U.S. Food and Drug Administration (FDA) approval. Some tool compounds of Smo, agonists such as purmorphamine, have demonstrated an ability to promote human embryonic stem cell differentiation (15), but the preclinical development of such small-molecule Smo agonists has lagged.

The interaction of Hedgehog ligand with the membrane protein Patched (Ptc) enables the seven-transmembrane receptor Smo to activate downstream Gli transcription factors (1). Activated Smo shares important behaviors with canonical G protein-coupled receptors (GPCRs), including an ability to undergo GPCR kinase phosphorylation and to recruit β -arrestin2 (β arr2) proteins for endocytosis, as shown in our previous study (16). Cyclopamine, a naturally occurring steroid alkaloid, inhibits the constitutive activity of Smo via direct antagonism, preventing its phosphorylation and interaction with β arr2. We exploited this observation to construct high-throughput high-content screens for Smo ligands, with a goal of accelerating the development of Hedgehog agonist drugs that could potentially have a role in tissue regeneration or be employed as tool compounds to study stem cell proliferation. We have identified four fluorinated glucocorticoids, halcinonide, fluticasone, clobetasol, and fluocinonide (Fig. 1A), all FDA-approved compounds, as Smo agonists that activate Hedgehog signaling and promote the proliferation

of primary neuronal stem/precursor cells alone and synergistically in the presence of Sonic Hedgehog (Shh).

Results

Fluorinated Glucocorticoid Smo Agonist (FGSA) Drugs Promote Smo Intracellular Aggregation with β arr2-GFP and Promote Smo Internalization. Included among the chemical libraries we screened using a 384-multiwell format was the Prestwick Chemicals Library containing FDA-approved drugs. It contains 68 glucocorticoids or structurally related steroid compounds, including cortisone and dexamethasone. The primary screening assay employed U2OS cells, chosen for adherence, flatness, and stable expression of β arr2-GFP, and a tail substitution mutant of Smo, Smo-633, which provided better sensitivity than WT Smo (16, 17). Images of Smo-633/ β arr2-GFP complexes were obtained at the rate of 5,000 per day using an automated confocal-based plate reader (ImageXpress Ultra; Molecular Devices). A read-out of compound activity for each well was provided by analyzing the corresponding image for changes in β arr2-GFP distribution that occurred as a result of compound addition.

When expressed in cells without the addition of exogenous Smo, β arr2-GFP is distributed homogeneously throughout the cytoplasm (Fig. 1B). The overexpression of Smo (16) or Smo-633 caused a redistribution of β arr2-GFP to intracellular vesicles/aggregates (Fig. 1C). At a concentration of 100 nM or greater, the Smo antagonist cyclopamine (18) reverses this effect and forces β arr2-GFP back into a homogeneous distribution (Fig. 1D). Intravesicular aggregation of β arr2-GFP can be restored in the presence of 100 nM cyclopamine with 5 μ M Smo agonist (SAG) (19) or purmorphamine (20, 21), both of which are known small-molecule Smo agonists (Fig. 1E and F). In our primary assay, a Smo agonist is identified by its ability to aggregate β arr2-GFP in the presence of 100 nM cyclopamine in the steady-state model.

Similar to the positive control, each hit compound at 5 μ M overcame the inhibition by 100 nM cyclopamine to produce intracellular β arr2-GFP aggregates (Fig. 1G–J). Hit compound agonist activities in U2OS cells were also confirmed using a β arr2-GFP assay with WT Smo (16) (Fig. S1). As assessed using the primary assay, the EC₅₀s for halcinonide, fluticasone, clobetasol, and fluocinonide are $1.1 \pm 0.1 \mu$ M, 99 ± 1.4 nM, $1.5 \pm 0.1 \mu$ M, and $>5 \mu$ M, respectively, whereas the EC₅₀s for the positive control agonists SAG and purmorphamine are 0.9 ± 0.1 nM and $>5 \mu$ M, respectively (Fig. 1K and Table 1). In com-

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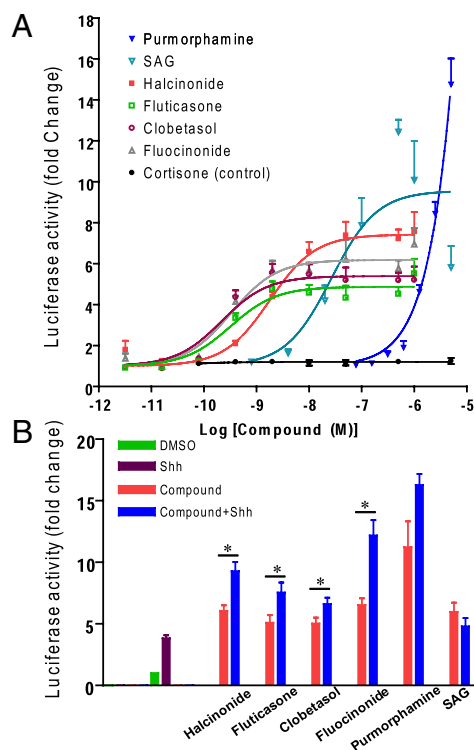


Fig. 4. Gli-luciferase response in Shh-LIGHT2 cells treated with Smo ligands. (A) Gli-luciferase reporter activity in Shh-LIGHT2 cells in response to Smo agonists. Shh-LIGHT2 cells cultured to confluence were individually treated for 30 h with the following compounds: halcinonide, fluticasone, clobetasol, fluocinonide, the positive controls purmorphamine and SAG, and the negative control cortisone. Results are presented as the mean $\pm$ SEM from multiple individual experiments ($n > 3$) performed in triplicate. (B) Effects of Shh-conditioned media (Shh) on Smo agonists. Shh-LIGHT2 cells were cultured to confluence and treated for 30 h with DMSO, 2% Shh, 5 μ M of the indicated compounds, or 5 μ M of the indicated compounds in the presence of 2% Shh. Results are presented as the mean $\pm$ SEM from multiple individual experiments ($n \geq 3$) performed in triplicate. The statistical significance was analyzed by a two-tailed Student's *t* test, with $*P < 0.05$ ($\alpha = 0.05$) defined as significant.

Using [3 H]thymidine incorporation, we further investigated the relationship between Shh and the Smo agonists on GCP proliferation. Shh (2%) induced a 17-fold increase of GCP proliferation as opposed to the marginal GCP proliferation response to 5 μ M halcinonide, fluticasone, clobetasol, or fluocinonide; 0.073 μ M purmorphamine; or 0.008 μ M SAG (Fig. 5B). Treatment of the GCP cells with 2% Shh and an agonist compound resulted in increased GCP proliferation ranging from 30-fold (fluocinonide) to 95-fold (SAG), indicating strong synergism (Fig. 5B). Interestingly, the glucocorticoid receptor (GR) agonist dexamethasone had a tendency to inhibit the activity of 2% Shh (Fig. 5B), as previously described (25). To verify further the opposite effects that the Smo agonists have on proliferation compared to dexamethasone, we treated GCP cells with Shh (2% and 20%) and various concentrations of halcinonide or dexamethasone in the absence or presence of 2% Shh. Synergism between 2% Shh and halcinonide was observed in the proliferation assay; such responses were equal to or greater than the stimulatory effects produced by 20% Shh (Fig. 5C, *Left*). In comparison, dexamethasone inhibited Shh-activated GCP proliferation in a dose-dependent manner (Fig. 5C, *Right*).

It has been reported that Hedgehog signaling promotes GCP proliferation through up-regulation of cyclin D2 and inhibition of proteasomal degradation of caspase-3, whereas dexamethasone

and several other GR agonists have the opposite effect by inhibiting GCP proliferation and not inhibiting GCP apoptosis (25). Although all the fluorinated glucocorticoid smoothened agonists from our study possess the ability, like dexamethasone, to activate GR as assessed by a GR-GFP nuclear translocation assay (Fig. S6), GCP treatment with Shh; purmorphamine; SAG; or fluorinated halcinonide, clobetasol, and fluticasone (but not the weaker Smo agonist fluocinonide) increased endogenous cyclin D2 protein expression and inhibited caspase-3 degradation (Fig. 6). No such growth-enabling responses were observed in GCPs treated with cortisone, dexamethasone, prednisolone, or corticosterone, observations consistent with recent reports (25–27) (Fig. 6). The opposite response of dexamethasone in the GCP proliferation assay and its similar response in the nuclear receptor assay compared to the FGSAs suggest that the signal for the GCP proliferative response is independent of glucocorticoid nuclear receptor signaling and is most probably attributable directly to activation of Smo.

In summary, the drugs halcinonide, fluticasone, clobetasol, and fluocinonide function as Smo agonists, having an ability to bind Smo, promote Smo internalization, activate Gli, and synergistically stimulate the proliferation of primary neuronal precursor cells.

Discussion

Regenerative medicine is an emerging frontier of medical therapy that holds the promise of curing currently untreatable diseases by harnessing the body's ability to replace damaged tissues (28). Such therapies might prompt autonomous tissue repair or facilitate the implantation of engineered tissue derived from progenitors or stem cells (28, 29). Hedgehog activators are prime candidates for therapeutics to initiate or modulate tissue self-repair. In this study, we used neuronal precursor/progenitor cells derived from mouse cerebella and modulated their proliferation using these Hedgehog activators. Even though the Hedgehog pathway was identified over two decades ago, there are still only a few available research compounds that can modulate it, such as SAG and purmorphamine, and, for a variety of reasons, no clinically available agonist drugs targeted to Hedgehog-related diseases exist.

Despite our findings of FGSAs that activate Smo, a search of PubMed indicates they have no apparent association with topical cancers. One of the four hit FGSAs active in the Hedgehog assays was fluticasone. It is reported on the GlaxoSmithKline drug label that fluticasone is well tolerated orally and can be administered i.v. in humans and that its s.c. and oral median lethal doses in mice and rats are greater than 1,000 mg/kg (30).

Glucocorticoids, including the fluorinated glucocorticoids, are used clinically for the treatment of asthma, inflammation, and skin disease or injury. Other glucocorticoids, including dexamethasone, prednisone, cortisone, and corticosterone, are used to treat premature infants and have been observed to cause neuronal apoptosis and to inhibit neuronal precursors of the cerebellar granule neuronal lineage in a mouse model (25, 26). On the basis that Shh exposure cannot overcome the effects of dexamethasone but can antagonize the effects of hydrocortisone, Heine and Rowitch (25) recommend that hydrocortisone be used as a replacement for dexamethasone in infants because of the reduced potential for neurotoxicity. We have demonstrated in vitro that fluticasone and the other fluorinated steroids can be used to expand neuronal precursor cell populations and potentiate the ability of 2% Shh stimulation. Our results with FGSAs suggest that some glucocorticoids may even be neuronally protective, but this requires further clinical investigation in this particular case. For regenerative medicine, we propose that FGSAs could be used immediately orally or i.v. on an acute or chronic basis for testing in disease models in which an increase in Gli signaling and consequent Hedgehog-mediated repair are desirable, such as neovascularization after myocardial infarctions

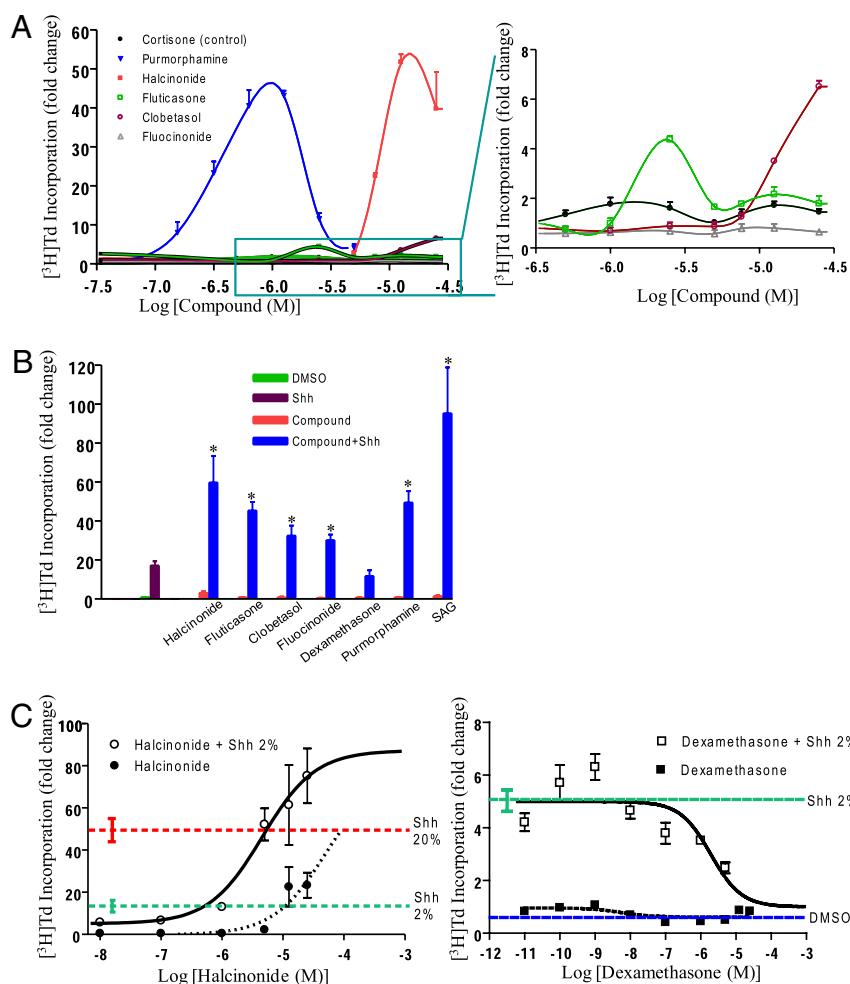


Fig. 5. Effects of FGSA and Shh on primary neuronal GCP proliferation. (A) Primary neuronal GCP proliferation data of Smo agonists. Expanded version (Right) of the boxed region (Left). Cells were treated with compounds for 48 h and then pulsed with [³H] thymidine ([³H]Td) and cultured for 16 h before being measured for [³H]Td incorporation. Cubic splines were fit to the data points using GraphPad Prism (GraphPad Software, Inc.) to highlight the responses. Data were acquired in triplicate from three independent experiments and are presented as the mean $\pm$ SEM. (B) Shh modulation of primary neuronal GCP proliferation in response to Smo agonists. The cells were treated with DMSO or 2% Shh alone or in the absence or presence of 2% Shh with one of the following compounds: 5 μ M halcinonide, fluticasone clobetasol, or fluocinonide; 5 μ M dexamethasone; and the positive control SAG (0.008 μ M) or purmorphamine (0.073 μ M). The [³H]Td incorporation data are presented as fold change vs. DMSO treatment, which was defined as 1. Triplicate data are presented as the mean $\pm$ SEM ($n = 3$). The statistical significance was analyzed by a two-tailed Student's t test, with $*P < 0.05$ ($\alpha = 0.05$) defined as significant (compound + Shh over Shh). (C) Halcinonide and dexamethasone have opposite effects on primary neuronal GCP proliferation. Cells were treated with DMSO, 2% Shh, 20% Shh, or halcinonide in the presence or absence of 2% Shh (Left) and with dexamethasone in the presence or absence of 2% Shh (Right; the minor change in responsiveness between experiments to 2% Shh treatment, may result from batch-to-batch variability in Shh). Dashed lines indicate the cell responses to DMSO vehicle, 2% Shh, and 20% Shh. Data acquired in triplicate are presented as the mean $\pm$ SEM ($n = 3$).

(10), wound healing in diabetes (13), or neuronal regeneration after spinal cord injury (12). In summary, the well-known pharmacokinetic and pharmacodynamic properties of these FDA-approved steroid Smo agonists provide a significant jumpstart in the process of beginning human studies on their potential therapeutic applications in regenerative medicine.

Materials and Methods

Materials. Details are described in *SI Materials and Methods*.

Transfection and Plasmids. Cells were transfected using either Fugene 6 (Roche) or Nucleofector (Amaxa). Details are provided in *SI Materials and Methods*.

Primary Assay-Automated High-Throughput Screening. We made multiple Smo mutants to identify the best location to attach the V2R tail (17), the addition of which, when phosphorylated and precisely located, causes β arr2-GFP to bind to Smo more strongly. Details are provided in *SI Materials and Methods*.

Bodipy-Cyclopamine Binding Analysis of Smo Agonists in Smo-Overexpressing HEK293 Cells. HEK293 cells stably expressing WT Smo were split at 166,000 cells per well in the center well (glass bottom, 10-mm diameter) of collagen-coated dishes (MatTek), followed by overnight incubation. Details are provided in *SI Materials and Methods*.

Gli-Luciferase Reporter Assay. The reporter assay using Shh-LIGHT2 cells, Smo^{-/-} mouse embryo fibroblasts (MEFs), or NIH 3T3 cells was performed as described (21).

[³H]Thymidine Proliferation Assay and Western Blots of Primary Neuronal GCPs. Primary GCPs were isolated from 4- or 8-day postnatal WT C57BL/6 mice, as previously described (23, 25). Details are provided in *SI Materials and Methods*.

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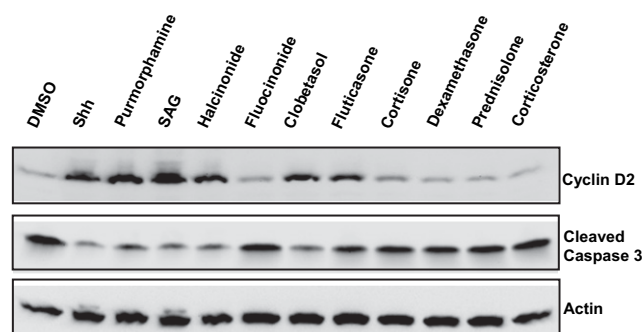


Fig. 6. Halcinonide, fluticasone, clobetasol, fluocinonide, and other glucocorticoids regulate cyclin D2 expression and caspase-3 degradation in primary neuronal GCPs. Primary neuronal GCPs derived from 4-day-old mice were individually treated for 64 h with DMSO, 2% Shh, 0.625 μ M purmorphamine, 0.5 μ M SAG, 2.5 μ M fluticasone, and the remaining compounds at 25 μ M. Cells were harvested in SDS sample buffer, protein samples were resolved on SDS/PAGE gels, and the corresponding immunoblots were probed by antibodies against cyclin D2, cleaved caspase-3, and actin ($n = 3$). A representative immunoblot is shown.

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